Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Forward Deployed Engineer building **production agentic AI and RAG systems** with Python, Spark, AWS, Azure, and GCP.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Translated customer requirements into a multi-tenant AI platform connecting FastAPI services, PostgreSQL, SQL Server, and cloud storage, running 5K+ monthly workflows with **99%+ production reliability** and documented operations.
- Architected asynchronous LLM orchestration with explicit job states, retries, and failure isolation, tripling throughput for 1K+ concurrent jobs while improving **scalability, observability, and deployment resilience** across enterprise workflows.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built AI-enabled React and Flask workflows supporting 150+ users and 10K+ daily requests, integrating GPT-4o, Claude, and Qwen-VL through **provider routing, validation, and automated fallbacks** in production.
- Designed concurrent Python pipelines for secure document ingestion, OCR, and multimodal processing, handling 1K+ files per batch with **automated testing**, failure recovery, and reliable CI/CD-driven cloud deployments.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python and PyTorch pipelines processing 100K+ autonomous-driving logs, improving planning 18% through **large-scale data engineering**, feature preparation, simulation evaluation, and documented analytical results for research stakeholders.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **model optimization**, debugging, failure analysis, and collaborative iterative technical reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered distributed integrations and ETL workflows using AWS Glue, S3, REST APIs, and object-oriented patterns, improving pipeline efficiency 68%, response times 30%, and **cross-platform system reliability** across environments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code reviews**, root-cause debugging, Git collaboration, and dependable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **secure REST APIs**, WebSockets, validation controls, and responsive React interfaces across distributed locations.
- Developed TypeScript integrations and real-time workflows, while Jenkins CI/CD, Docker staging, and **automated regression testing** reduced deployment time 50% and strengthened production readiness across controlled cloud releases.

Skills

AI & Agents: LangGraph, Semantic Kernel, AutoGen, CrewAI, Hugging Face, PyTorch, TensorFlow, RAG, Multi-Agent Systems, Vector Search
Languages & Data: Python, TypeScript, C#, SQL, Pandas, Spark, PostgreSQL, MongoDB, Redis, ETL, Data Modeling
Cloud & Systems: AWS, Azure, GCP, Azure OpenAI, Docker, Kubernetes, Linux, Distributed Systems, REST APIs, FastAPI, Microservices, CI/CD
Delivery: Git, GitHub, PyTest, Integration Testing, Code Review, Debugging, Performance Optimization, Security, Observability, Agile, SDLC, Documentation, Requirements Analysis, Stakeholder Communication

Projects

Enterprise Agent Operations Copilot | *Python, LangGraph, Azure OpenAI, Spark, Kubernetes*

- Designed a governed multi-agent system that diagnoses operational incidents, retrieves enterprise knowledge, proposes remediation, and executes approved actions through **policy guardrails, human checkpoints, auditable traces**, observability, and Kubernetes deployment.

Agentic Modernization Studio | *Python, AutoGen, Hugging Face, AWS, Terraform, GitHub Actions*

- Built an agent orchestration workflow that analyzes legacy applications, plans modernization tasks, generates reviewed changes, and captures deployment learnings through **specialized agent coordination**, automated evaluations, secure tooling, and cloud CI/CD.

Big-Brain | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a retrieval-augmented generation platform combining LLaMA-2, FAISS, Pinecone, PostgreSQL, and secure REST APIs, deploying **structured and unstructured knowledge retrieval** on GCP for reliable contextual enterprise responses.